PROJECT PROPOSAL

FraudGuard — Financial Fraud Detection & Customer Risk Analysis

Omkar Padhi — 24101C0003 **Shremaan Kumar — 24101C0002**

Data Warehouse and Mining (DWM) · Department of INFT · Vidyalankar Institute of Technology

Academic Year 2026–2027 **Project Guide:** Shruti Agrawal

1. Introduction

Lenders process many loan applications, a small share of which are fraudulent. FraudGuard is a web-based system that applies Data Warehouse and Mining to this problem: a star schema and OLAP to explore historical loan data along several dimensions, and classification models to learn fraud patterns from labelled history.

2. Problem Statement

Manual loan review is slow and subjective, records no reason for decisions and gives no view of fraud across time, loan types or customer segments. Fraud is rare, so simple rules and accuracy-based evaluation are not adequate.

3. Objectives

(i) Model the loan data as a star schema and analyse it with OLAP; (ii) assess data quality and perform EDA; (iii) build an ML pipeline that handles class imbalance and compare five classifiers; (iv) provide Customer, Bank Officer and Admin / Technical modules with role-based access and an audit trail.

4. Proposed Methodology

Data profiling → star schema design → preprocessing (cleaning, imputation, scaling, encoding) → stratified 80 / 20 split with SMOTE on training data → train and evaluate five models → backend API, three modules and testing. Tech: React, TypeScript, FastAPI, pandas, scikit-learn, XGBoost, SQLite.

5. Dataset

5,000 historical loan applications with 25 attributes (5 June 2022 to 4 June 2025); target FraudFlag with 103 fraud (2.06%) and 4,897 genuine records. No credit score is available.

6. DWM Concepts

Fact table FACT_LOAN_APPLICATION with Date, Customer, Loan, Employment and Fraud Status dimensions; roll-up, drill-down (Year → Quarter → Month → Loan Type), slice, dice, pivot; ETL and data-quality checks.

7. ML Approach

Binary classification on 13 features with Logistic Regression, Decision Tree, Naive Bayes, Random Forest and XGBoost; evaluated by accuracy, precision, recall, F1, ROC-AUC and confusion matrix. The fraud probability is shown to the officer as a risk level; the officer decides.

8. System Modules

Customer: apply for a loan, track status and rejection reason. **Bank Officer:** review applications with fraud probability; approve, reject or send for review. **Admin / Technical:** data quality, model comparison, OLAP and system information.

9. Expected Outcomes

OLAP analysis, a measured comparison of five models, and a working three-role web application with audit trail; results reported as measured.

10. Scope

In scope: given dataset, model comparison, loan workflow. **Out of scope:** document verification, bank integration, production deployment.

11. Timeline

Wk 1–4: proposal, dataset · Wk 5–6: star schema, EDA, OLAP · Wk 7–8: ML models · Wk 9–12: backend and modules · Wk 13–14: testing, report.

12. References

[1] Han, Kamber, Pei, Data Mining: Concepts and Techniques, 2011. [2] Kimball, Ross, The Data Warehouse Toolkit, 2013. [3] Chawla et al., “SMOTE,” JAIR, 2002. [4] Chen, Guestrin, “XGBoost,” ACM SIGKDD, 2016.

Omkar Padhi	Shremaan Kumar	Shruti Agrawal (Project Guide)